

No-Family-Indicator v1: Candidate Supplement (Non-Submission)

Isolated H001 evaluation artifact

1 Scope

This candidate removes only the constant family indicator from each family-specific compatibility head. All splits, targets, normalization, optimization, ranking, metrics, and uncertainty procedures are unchanged. Internal development is used only as a sanity check. The active manuscript is not modified by this artifact.

2 Official-validation results

3 Controls and construct checks

4 Artifact lock

The candidate model, score contract, protocol files, compact evaluation summaries, and SHA256 manifest are included in the accompanying release candidate.

Predictor	Ranking rule	$K = 5$		$K = 10$		$K = 20$		$K = 50$		$K = 100$	
		R	V	R	V	R	V	R	V	R	V
Open3DSG	Source	3.42	52.05	9.87	32.89	19.89	20.99	40.43	13.87	51.11	12.42
	RelCompat3D	3.73	0.94	11.38	2.33	23.62	3.13	44.18	3.42	56.85	3.24
	Matched MLP	3.70	4.95	11.78	4.97	24.67	4.56	46.70	4.13	59.89	3.71
	RankAvg	3.78	1.13	11.05	2.51	22.36	3.55	43.20	4.60	54.00	5.49
	RRF	3.58	19.59	10.67	16.47	21.58	13.60	42.45	9.60	54.68	7.80
	Product (all families)	9.94	3.98	19.64	4.54	32.80	4.37	46.42	2.69	60.05	3.31
VL-SAT	Source	41.94	0.29	63.22	0.82	80.74	1.42	92.72	2.68	96.35	4.76
	RelCompat3D	42.07	0.15	63.39	0.57	80.82	1.14	92.77	1.97	96.58	2.95
	Matched MLP	42.09	0.15	63.47	0.51	80.92	1.09	92.72	1.89	96.50	2.96
	RankAvg	35.32	0.15	51.74	0.47	69.41	0.99	90.23	1.62	97.05	2.26
	RRF	36.81	0.15	55.66	0.51	75.08	1.07	91.72	1.82	97.05	2.50
	Product (all families)	41.74	0.11	63.32	0.49	80.94	1.10	92.93	2.03	96.88	3.25
SGFN	Source	31.17	2.37	39.75	3.49	49.12	3.22	74.02	3.85	92.35	6.30
	RelCompat3D	31.17	2.37	39.75	3.47	49.14	2.97	74.50	2.63	93.03	3.50
	Matched MLP	31.17	2.37	39.75	3.47	49.19	2.96	74.57	2.58	92.88	3.50
	RankAvg	31.09	2.37	38.80	3.47	46.63	2.95	70.22	2.35	92.72	2.60
	RRF	31.14	2.37	38.95	3.47	47.08	2.97	68.50	2.54	87.99	3.07
	Product (all families)	31.07	2.48	40.56	3.19	51.18	2.76	77.04	2.58	94.13	3.71

Table 1: Candidate no-family-indicator results. Exact-label Recall ($R\uparrow$) and verifier-derived Violation ($V\downarrow$) are percentages. All matched methods use the unchanged family-aware ranking procedure.

Predictor	K	Δ Recall	95% CI	Δ Violation	95% CI
VL-SAT	5	+0.13	[+0.00,+0.26]	-0.15	[-0.30,-0.03]
	10	+0.18	[+0.05,+0.35]	-0.26	[-0.45,-0.09]
	20	+0.08	[-0.03,+0.19]	-0.28	[-0.43,-0.16]
	50	+0.05	[-0.05,+0.18]	-0.70	[-0.88,-0.55]
	100	+0.23	[+0.03,+0.46]	-1.82	[-2.00,-1.65]
Open3DSG	5	+0.30	[+0.11,+0.49]	-51.11	[-54.39,-47.86]
	10	+1.51	[+1.04,+2.00]	-30.56	[-32.55,-28.51]
	20	+3.73	[+2.96,+4.46]	-17.86	[-18.97,-16.74]
	50	+3.75	[+2.96,+4.50]	-10.45	[-11.06,-9.86]
	100	+5.74	[+4.63,+6.86]	-9.18	[-9.62,-8.69]
SGFN	5	+0.00	[+0.00,+0.00]	+0.00	[+0.00,+0.00]
	10	+0.00	[+0.00,+0.00]	-0.02	[-0.06,+0.00]
	20	+0.03	[-0.07,+0.14]	-0.25	[-0.43,-0.11]
	50	+0.48	[+0.29,+0.69]	-1.22	[-1.44,-1.01]
	100	+0.68	[+0.42,+0.98]	-2.80	[-3.05,-2.54]

Table 2: Paired scan-cluster intervals for RelCompat3D minus Source at every reported budget. Entries are percentage points.

Predictor	Condition	R@50	V@50	R@100	V@100
Open3DSG	RelCompat3D	44.18	3.42	56.85	3.24
	Wrong predicate	42.65	21.98	53.47	20.96
	Wrong pair	38.52	8.50	49.32	8.36
	Shuffled geometry	38.42	12.93	48.79	12.42
	Fixed-predicate swap	42.67	21.98	53.63	20.96
	Distance only	51.16	8.24	63.22	9.55
	Compatibility only	42.22	3.42	56.87	3.24
VL-SAT	RelCompat3D	92.77	1.97	96.58	2.95
	Wrong predicate	90.46	4.97	94.79	7.95
	Wrong pair	90.56	2.61	94.96	4.73
	Shuffled geometry	89.17	3.05	94.11	5.60
	Fixed-predicate swap	90.46	4.97	94.81	7.95
	Distance only	81.90	5.34	89.80	8.09
	Compatibility only	67.65	1.61	83.89	2.03
SGFN	RelCompat3D	74.50	2.63	93.03	3.50
	Wrong predicate	71.58	9.42	89.93	12.99
	Wrong pair	71.55	3.99	88.07	6.65
	Shuffled geometry	70.59	4.73	86.83	8.36
	Fixed-predicate swap	71.58	9.42	90.03	12.98
	Distance only	63.19	10.00	84.06	12.66
	Compatibility only	52.79	2.32	70.64	2.30

Table 3: Candidate ablations under the same candidates and ranking procedure. All entries are percentages.

Predictor	Condition	R@50	V@50	R@100	V@100
Open3DSG	Family product	46.85	2.90	60.52	3.40
	Pooled product	47.18	5.28	64.43	7.47
	Hard-rule filter	42.95	0.00	53.68	0.00
VL-SAT	Family product	92.93	2.03	96.88	3.25
	Pooled product	93.00	2.19	96.90	3.87
	Hard-rule filter	92.57	0.00	96.27	0.00
SGFN	Family product	77.04	2.58	94.13	3.71
	Pooled product	79.25	2.92	94.13	4.64
	Hard-rule filter	74.62	0.00	92.70	0.00

Table 4: Unrestricted family-conditioning and hard-filter comparisons. Values are percentages; the hard filter may select fewer than K candidates.

Predictor	Ranking	V_{all}	V_{dec}	U	V_{pess}
VL-SAT	Source	0.0476	0.0729	0.3464	0.3941
	RelCompat3D	0.0295	0.0452	0.3473	0.3768
Open3DSG	Source	0.1242	0.2126	0.4161	0.5402
	RelCompat3D	0.0324	0.0548	0.4085	0.4409
SGFN	Source	0.0630	0.1005	0.3732	0.4362
	RelCompat3D	0.0350	0.0565	0.3803	0.4153

Table 5: Verifier-uncertainty sensitivity at $K = 100$.

Predictor	Source	Ours	Change (95% CI)	Coverage
VL-SAT	16.43	14.34	-2.09 [-2.41, -1.80]	95.5/71.0
Open3DSG	45.96	4.90	-41.06 [-42.75, -39.28]	98.2/83.7
SGFN	11.56	8.13	-3.43 [-3.92, -3.00]	95.8/79.9

Table 6: Candidate surface-based Violation at $K = 50$ using point-mesh consensus. Coverage is measured/decidable.

Predictor	Condition	K=50		K=100		$\Delta R@50$	$\Delta V@50$
		R	V	R	V		
VL-SAT	Main	92.77	1.97	96.58	2.95	+0.00	+0.00
	Remove exact scalar	92.80	1.99	96.60	3.00	+0.03	+0.01
	Remove measurement family	92.75	2.68	96.53	4.68	-0.03	+0.70
	Alternative evidence only	92.80	2.68	96.42	4.71	+0.03	+0.70
Open3DSG	Main	44.18	3.42	56.85	3.24	+0.00	+0.00
	Remove exact scalar	43.81	3.42	56.87	3.25	-0.38	+0.00
	Remove measurement family	41.19	9.54	52.64	9.55	-3.00	+6.12
	Alternative evidence only	41.04	11.94	53.27	11.88	-3.15	+8.51
SGFN	Main	74.50	2.63	93.03	3.50	+0.00	+0.00
	Remove exact scalar	74.65	2.68	93.00	3.58	+0.15	+0.04
	Remove measurement family	74.67	3.88	92.98	6.16	+0.18	+1.24
	Alternative evidence only	74.62	3.86	92.77	6.25	+0.13	+1.23

Table 7: Candidate feature-removal analysis. Deltas compare each condition with the no-family-indicator main route.

Condition	Dev order	VL-SAT K=50		Open3DSG K=50		SGFN K=50	
		R	V	R	V	R	V
default	100.00	92.77	1.97	44.18	3.42	74.50	2.63
negative_cap_1	100.00	92.77	1.97	44.28	3.42	74.47	2.63
negative_cap_4	100.00	92.77	1.97	44.16	3.42	74.55	2.63
pair_weight_0.125	100.00	92.77	1.98	44.21	3.42	74.50	2.64
pair_weight_0.5	100.00	92.77	1.97	44.21	3.42	74.50	2.62
proximity_2.0	100.00	92.80	1.97	44.36	3.42	74.70	2.63
proximity_3.0	100.00	92.75	1.98	44.26	3.42	74.37	2.64
vertical_0.20m	100.00	92.77	1.94	44.11	3.42	74.50	2.58
vertical_0.30m	100.00	92.77	2.04	44.39	3.42	74.47	2.74

Table 8: One-factor counterfactual-policy sensitivity after removing the constant family indicator. No condition is selected on validation.